

GenAI Project Reflection Report

1. What I Learned About Generative AI

In this project, I learned that Generative AI, also known as GenAI, is a type of artificial intelligence that can create new content such as text, images, code, summaries, and explanations. Tools such as ChatGPT are examples of GenAI systems because they can generate responses based on the prompts users provide.

One important concept I learned is that GenAI models are trained on large amounts of data.

These models use patterns from the data to generate new responses. For example, when AI generates text, it predicts the next word or phrase based on the words that came before it. This allows the system to create sentences that sound natural and organized.

For example, when I gave the AI a prompt such as, “Write a summary about tourism,” it generated a structured paragraph automatically. This showed me how GenAI can help with writing tasks, brainstorming, and organizing ideas.

I also learned that prompts are very important when using GenAI. A weak prompt usually gives a general answer, while a strong prompt gives a more specific and useful response.

For example:

Weak prompt: “Write something about business.”

Strong prompt: “Write a 150-word summary about marketing strategies for small businesses.”

This helped me understand that using GenAI effectively requires clear instructions, enough context, and careful review of the output.

2. What I Learned About Building an AI System

I learned that building an AI system involves several important steps. First, data collection is needed because AI systems depend on data to work properly. In this project, we used Excel or Google Sheets to organize information and results.

Second, data processing is important. Before using data, it needs to be cleaned, organized, and checked for errors. If the data is messy or incomplete, the AI system may produce weak or inaccurate results.

Third, I learned about model usage. Instead of building an AI model from the beginning, this project showed how existing AI tools, APIs, or libraries can be used to support a system. This made the process easier and helped me understand how AI tools can be applied in real projects. Fourth, testing is an important part of building an AI system. We tested the system by giving it different inputs and reviewing the outputs. This helped us see whether the system was working correctly.

Finally, improvement is necessary. At first, the AI gave very general answers. After improving the prompt and giving more specific instructions, the output became more detailed and useful. This showed me that AI systems often need multiple revisions before they work well.

3. Challenges Faced

During this project, I faced several challenges. One challenge was understanding how prompts affect the AI’s output. Small changes in wording can make the response more accurate or more confusing.

Another challenge was getting accurate results from the AI. I learned that AI can be helpful, but it still needs human review because it can sometimes give answers that are too general, incomplete, or incorrect.

I also learned that organizing code, data files, and project documents is important. A clear folder structure makes the project easier to understand and submit.

4. Conclusion

Overall, this project helped me understand both how Generative AI works and how AI systems are built in practice. I learned that GenAI can support writing, coding, summarizing, and problem-solving, but it works best when users provide clear prompts and review the results carefully.

This project also taught me that building an AI system requires planning, data organization, testing, and improvement. I now have a better understanding of how to use AI tools effectively and how to structure an AI-based project in a professional way.